Karthik Raja Subramanian

United States | +17819216685 | karthiks233@gmail.com | Linkedin | Github

WORK EXPERIENCE

Gold Star Foods **LA, CA**

Software Engineering *Jan 2026 - Present*

- Delivered full-stack features powering procurement and demand-forecasting workflows on an agentic food distribution platform, as measured by 5+ features shipped end-to-end, by building across TypeScript, React, Express, Prisma, and AWS CDK.
- Accelerated a bidirectional Text-to-SQL chatbot, as measured by a 40% drop in query-generation latency, by building on AWS Bedrock's Converse API with tool-use calls and knowledge-graph prompt injection while enforcing guardrails at the prompt and query-validation layers.
- Automated the dev deployment path, as measured by deploy time, cut from 30 to 5 min zero-touch deploys per week, by designing a GitHub Actions CI/CD pipeline with build, lint, and TypeScript type-check gates plus gated manual approval for production.
- Reduced new-developer onboarding time, as measured by an estimated 30% decrease, by authoring production runbooks for 5+ microservices across Docker, PostgreSQL, and AWS ECS and standardizing incident response across environments using efficient bug tracking protocol and alert system.
- Administered the platform for production observability, as measured by 60% faster incident detection / 10 dashboards & alarms, by wiring CloudWatch & Sentry metrics along with structured logs, and alarms across services.

Homesite (American Family Insurance) **Boston, MA**

Cloud Engineer Intern *May 2024 - Dec 2024*

- Maintained production reliability for real-time analytics and batch-processing pipelines, as measured by 99.9% uptime, by defining technical deployment requirements for cloud workloads.
- Strengthened security posture and reduced workflow ambiguity, as measured by a 15% drop in ambiguity, by orchestrating cloud resource allocation and least-privilege IAM policies for engineering squads.
- Improved team delivery efficiency, as measured by a 20% increase, by leading CI/CD and Infrastructure-as-Code demos that drove self-service tooling adoption across squads.
- Aligned product backlogs with platform scalability goals, as measured by 20 backlogs / 75% sprint-predictability gain, by partnering with SRE teams in JIRA-driven Agile ceremonies.
- Minimised cloud spend, as measured by 20% reduction, by right-sizing compute and storage and flagging idle resources during recurring cost reviews

Tata Consultancy Services (TCS) **Bangalore, India**

Systems Engineer *Aug 2021 - Aug 2023*

- Maintained business-critical telecom services, as measured by 99.9% uptime across 500+ virtualized nodes, by administering production Linux (Amazon Linux, RHEL) environments.
- Eliminated operational toil, as measured by a 30% reduction, by developing Python and Shell self-service tooling for log ingestion, system patching, and routine maintenance that let teams self-resolve.
- Lowered infrastructure-incident MTTR, as measured by 70% reduction, by engineering log-parsing agents that auto-detected faults across distributed networking devices.
- Increased successful production-deployment rate, as measured by a 15% gain, by optimizing the Agile release pipeline with JIRA automation and Git version control.
- Streamlined incident handling, as measured by 30% fewer escalations, by authoring standardized runbooks and troubleshooting guides for the on-call team.

EDUCATION

Northeastern University **Boston, MA**

Master of Science in Software Engineering *Graduation Date: Dec 2025*

CMR University **Bangalore, India**

Bachelor of Engineering in Electronics & Communication *Graduation Date: Aug 2021*

PROJECT EXPERIENCE

Cloud-Native AI Infrastructure Automation **Boston, MA**

Self Project

- Achieved configuration consistency across Dev, Staging, and Production, as measured by 100% parity, by provisioning highly available AWS infrastructure (VPC, EKS, S3) as code with Terraform.
- Cut deployment lead time and eliminated infrastructure drift, as measured by a 70% reduction in lead time, by implementing a GitOps CI/CD pipeline in Jenkins with automated testing and Terraform state locking.
- Scaled application workloads, as measured by 5 services / 60% faster deploys, by designing Docker and Kubernetes (EKS) orchestration workflows for build, test, and deploy.

GitOps & Secrets Platform (Homelab) **Boston, MA**

Self Project

- Standardized multi-environment Kubernetes deployments, as measured by [50–70%] less manifest duplication across Dev/Staging/Prod, by templating workloads into reusable Helm charts with per-environment values files.
- Automated continuous, self-healing delivery, as measured by automatic drift correction (every 3 min, near-instant via webhook) and rollback to the last healthy revision in under 30s, by migrating from a push pipeline to ArgoCD pull-based GitOps.
- Hardened pipeline secrets management, as measured by 100% of static credentials replaced with short-lived dynamic secrets (TTL [15m–1h]), by integrating HashiCorp Vault with least-privilege policies and EKS workload identity (IRSA).
- Shifted security left in CI, as measured by 100% of pull requests gated by automated secret and code scanning, by adding Gitleaks and Semgrep checks to the GitHub Actions pipeline.

SKILLS & INTERESTS

Cloud: AWS (EKS, EC2, S3, IAM, VPC, CloudWatch), Google Cloud (GKE)

CI/CD & GitOps: GitHub Actions, Jenkins, ArgoCD, Helm, GitOps workflows

IaC & Config:: Terraform, Ansible

Containers & Orchestration: Docker, Kubernetes (EKS), Helm

Security & Secrets: AM (least-privilege), HashiCorp Vault, secret/code scanning (Gitleaks, Semgrep), OIDC/SAML

Observability: Prometheus, Grafana, Sentry, CloudWatch

Networking: TCP/IP, DNS, DHCP, VPC, load balancers, NAT

Languages: Python, Bash, PowerShell, Java, C, SQL

Operating Systems: Linux (Amazon Linux, Ubuntu, RHEL), Windows Server

Databases: MySQL, PostgreSQL

AI-assisted Dev: AWS Bedrock, agentic tooling

Certification: AWS Certified Solutions Architect – Associate (Credential ID: b83b7ae2536e415ca1dbf352fa7cb112)